

Causal Inference Powered by Hybrid LLM Synthetic Data Generation

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1 Introduction

The increasing capabilities of Large Language Models (LLMs) offer exciting opportunities to develop AI agents that can assist with complex analytical workflows. This project focuses on constructing an LLM agent specifically designed to power and facilitate reliable causal inference, not by performing the causal estimation directly, but by addressing key data-related challenges inherent in causal analysis. The core function of this "LLM agent for causal inference" is to generate high-fidelity synthetic data that is uniquely suited for downstream causal analysis using traditional estimators. This approach is crucial for overcoming limitations of real-world data such as scarcity, privacy constraints, and the difficulty in obtaining datasets with complete covariate balance or sufficient overlap to satisfy causal assumptions. The primary challenge lies in enabling the agent to generate synthetic data that not only mimics the observational characteristics of real data but, critically, preserves the underlying causal structures required for accurate estimation of causal effects. Furthermore, for the synthetic data produced by the agent to truly enable reliable inference, it must help mitigate common methodological issues in causal analysis, such as violations of the positivity assumption, and provide a robust basis for evaluating the performance of the causal estimators that will be applied to it. Our work directly tackles these foundational data generation and evaluation challenges, providing essential capabilities for an LLM agent focused on empowering reliable causal inference through synthetic data.

Our main contributions, therefore, directly support the data generation and utility functions of an LLM agent for causal inference:

- We observe that existing generative models for privacy-preserving data sharing often fail to preserve causal parameters, even when they maintain strong predictive performance.
- We propose a hybrid data generation approach designed to preserve causal structure, particularly the average treatment effect (ATE).
- We introduce an algorithm that leverages synthetic data to mitigate violations of the positivity assumption in causal inference.
- We propose a method for evaluating the performance of causal estimators in scenarios where the ground truth is unknown in the real dataset.

2 Related works

Seldom work investigate the intersection of synthetic data and causal inference.

3 Using LLMs as Realistic Synthetic Data Generators

In this section, we describe how we employed a Large Language Model (LLM), specifically GPT-2, to generate realistic synthetic tabular data using the GReaT framework Borisov et al. (2023). This procedure underpins our hybrid data-generation strategy, enabling the accurate simulation of covariate-treatment-outcome triplets suitable for causal inference analysis.

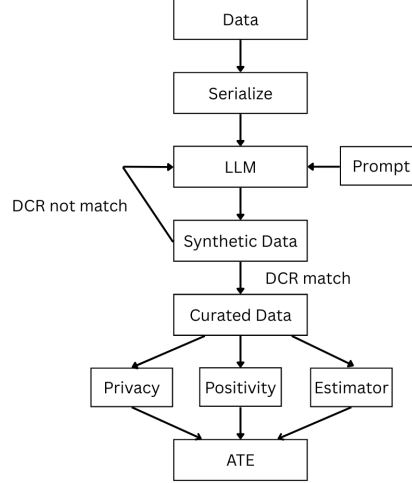


Figure 1: Overview of the LLM-based synthetic data generation pipeline. Raw data is first serialized into prompts and processed through an LLM. The resulting synthetic samples are filtered via DCR (distance to closest record) matching. Curated synthetic data is evaluated for privacy preservation, positivity balance, and estimator fidelity before contributing to ATE estimation.

3.1 Textual Serialization and Prompt Construction

We serialized each row of the tabular dataset into natural language sentences following the methodology proposed in the GReaT framework. Each covariate, treatment, and outcome was encoded explicitly as a subject-predicate-object clause, structured as follows:

$$W1=1, W2=0, W3=1, W4=0, W5=1, W6=0 \Rightarrow A=1, Y=0$$

This textual encoding preserves the semantics of each feature, enabling the LLM to model complex dependencies among covariates, treatments, and outcomes without imposing artificial ordering biases. To enhance generalizability and allow conditioning on arbitrary feature subsets, we randomized the order of serialized features prior to training.

3.2 Fine-Tuning with GReaT

We fine-tuned GPT-2 using the GReaT Python package Borisov et al. (2023), which facilitates the application of pretrained language models for generating synthetic tabular data via textual encodings. We conducted training for 50 epochs with a batch size of 32, configuring the model to save checkpoints every 400,000 training steps using the `save_steps` parameter to ensure reproducibility and enable checkpoint recovery. Our training dataset comprised 5,740 samples, each containing six binary covariates (W), a binary treatment (A), and a binary outcome (Y). The fine-tuning configuration was as follows:

- **Model:** GPT-2 (355M parameters)
- **Epochs:** 50
- **Batch size:** 32
- **Synthetic sample count:** 50,000
- **Maximum sampling length:** 1024–2000 tokens

After completing the training phase, we saved and subsequently reloaded the trained model to generate synthetic data. We exported the generated data in a structured tabular format to facilitate downstream causal analysis. All intermediate and final outputs follow a systematic file-naming convention to enhance traceability and reproducibility and are accessible in the supplementary repository.

3.3 Sampling Strategy

We leveraged the autoregressive capabilities of GPT-2 to generate synthetic records sequentially, conditioned on randomized feature orders. We conducted sampling with a maximum generation length ranging from 1024 to 2000 tokens. The flexibility inherent in the GReaT framework permitted both complete record generation and conditional sampling based on subsets of features, thereby facilitating advanced applications such as counterfactual simulations and covariate imputations.

3.4 Advantages and Fidelity Considerations

The GReaT framework supports arbitrary conditioning and obviates the need for lossy preprocessing methods, such as one-hot encoding. This feature is particularly beneficial for causal inference tasks, where maintaining complex conditional dependencies is critical. Furthermore, distance-to-closest-record (DCR) analysis confirmed that the generated samples closely matched the original data distribution, while simultaneously preserving sufficient variability to prevent overfitting and data leakage.

3.5 Runtime and Reproducibility

Across multiple runs, fine-tuning and sampling procedures averaged between 15 and 20 minutes, depending on system specifications. Our pipeline ensures reproducibility through comprehensive storage of model checkpoints and deterministic sampling configurations.

4 Failure of Generative Model to Preserve Causal Parameter and A Hybrid generation approach

Generative model-based data encryption using synthetic data generation to protect privacy has gained increasing attention, with approaches based on GANs Xu et al. (2019) and LLMs Borisov et al. (2023). While metrics such as train-on-synthetic-test-on-real (TSTR) are commonly used to evaluate the predictive quality of synthetic data, there is limited focus on whether these data preserve causal parameters. The most commonly studied causal parameter is the average treatment effect (ATE), denoted as $\mathbb{E}[Y^1 - Y^0]$.

To preserve the causal parameters of a dataset beyond mere predictive performance, we focus on the tuple (W, A, Y) , where W denotes covariates, A is a binary treatment, and Y is a binary outcome. Our goal is to assess whether generative models can preserve underlying causal relationships, not just surface-level predictive associations. Under standard causal assumptions (e.g., ignorability, positivity, and consistency), the average treatment effect (ATE) is identified as

$$\Psi(P) = \mathbb{E}_P[\mathbb{E}_P[Y \mid A = 1, W] - \mathbb{E}_P[Y \mid A = 0, W]].$$

Crucially, this identification relies on how Y varies with A conditional on W . Therefore, accurate modeling of the conditional distribution $Y \mid A, W$ does not necessarily imply accurate recovery of the causal parameter.

Inspired by this, we propose a hybrid data generation approach to address the challenge that generative AI may fail to preserve causal parameters. As shown in Algorithm 1, we begin by training a generative model on the seed dataset $\mathcal{D}_{\text{seed}}$ to produce synthetic covariates $\tilde{W}$, using the distance to closest records (DCR) as a monitoring metric. Next, we fit a propensity score model $\hat{g}(A \mid W)$ and an outcome model $\hat{h}(Y \mid A, W)$ using a predictive algorithm such as random forest, both trained on $\mathcal{D}_{\text{seed}}$. Finally, for each synthetic covariate $\tilde{W}$, we sample a treatment $\tilde{A}$ from $\hat{g}$ and generate an outcome $\tilde{Y}$ from $\hat{h}$, thereby forming the synthetic triplet $(\tilde{W}, \tilde{A}, \tilde{Y})$.

Algorithm 1 Hybrid Synthetic Data Generation Preserving Causal Structure

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1: Input: Seed dataset  $\mathcal{D}_{\text{seed}} = \{(W_i, A_i, Y_i)\}_{i=1}^n$ 
2: Train:
   - Train generative model  $p_\theta(W)$  on  $\{W_i\}_{i=1}^n$ , using DCR as a monitoring metric
   - Propensity model  $\hat{g}(A \mid W)$  on  $\mathcal{D}_{\text{seed}}$ 
   - Outcome model  $\hat{h}(Y \mid A, W)$  on  $\mathcal{D}_{\text{seed}}$ 
3: for  $i = 1$  to  $n$  do
4:   Sample  $\tilde{W}_i \sim p_\theta(W)$  ▷ Generate synthetic covariate
5:   Compute  $\hat{g}_i = \hat{g}(1 \mid \tilde{W}_i)$  ▷ Propensity score
6:   Sample  $u \sim \text{Uniform}(0, 1)$ 
7:   if  $u < \hat{g}_i$  then
8:     Set  $\tilde{A}_i = 1$ 
9:   else
10:    Set  $\tilde{A}_i = 0$ 
11:   end if
12:   Generate  $\tilde{Y}_i = \hat{h}(\tilde{A}_i, \tilde{W}_i)$ 
13: end for
14: Return: Synthetic dataset  $\{(\tilde{W}_i, \tilde{A}_i, \tilde{Y}_i)\}_{i=1}^n$ 

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We benchmark synthetic data generated by large language models (LLMs), generative adversarial networks (GANs), and their corresponding hybrid variants. Our findings reveal that current generative AI methods for tabular data synthesis can fail to preserve key causal parameters, such as the average treatment effect (ATE). In the fully synthetic setting, denoted as "Syn Full," both treatment $\tilde{A}$ and outcome $\tilde{Y}$ are generated directly from the generative model p_θ . To ensure privacy while maintaining similarity to the original data distribution, the synthetic covariates $\tilde{W}$ are generated with a distance to closest records (DCR) comparable to that of the real data, as shown in Figure 2. The true (theoretical) ATE in our setting is 0.418256.

As shown in Table 1, LLM-generated synthetic data achieves the highest TSTR performance among all synthetic data methods. However, the ATE estimates obtained using either IPTW or AIPW are close to 0.5, which is substantially higher than the true ATE. In contrast, the hybrid approach consistently yields significantly more accurate ATE estimates compared to the fully synthetic (Syn Full) method, for both GAN- and LLM-based models.

5 A valid way to address positivity with LLM synthetic data

We then propose using synthetic data to address violations of the positivity assumption. Specifically, we observe that pairing samples with extreme propensity scores with synthetic counterparts can significantly reduce bias in both IPTW and AIPW estimators. In contrast, standard truncation techniques offer little benefit in this setting, as samples with near-zero propensity scores consistently receive $A = 0$. This highlights

Table 1: AUC TSTR, IPTW and AIPW ATE Estimates, and Their Absolute Differences from Theoretical ATE

Baseline	AUC	IPTW ATE	AIPW ATE	IPTW Diff	AIPW Diff
Data Seed	0.7705	0.3959	0.3770	0.0223	0.0412
GAN Syn Hybrid	0.7425	0.3538	0.3525	0.0645	0.0657
GAN Syn Full	0.4583	-0.0378	-0.0318	0.4561	0.4501
LLM Syn Hybrid	0.7512	0.4295	0.4205	0.0112	0.0022
LLM Syn Full	0.7545	0.5067	0.4989	0.0884	0.0806

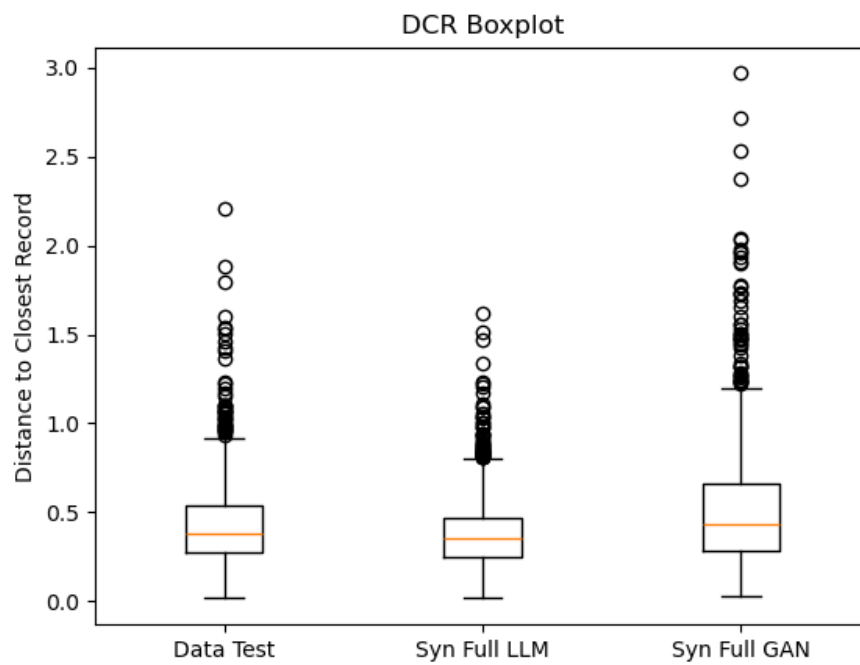


Figure 2: Boxplot of Distance to Closest Record (DCR).

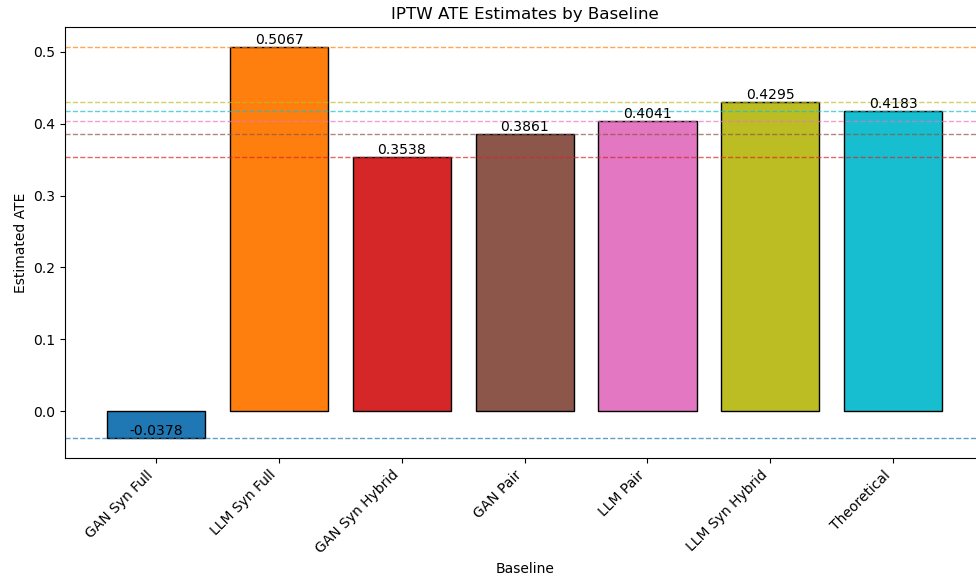


Figure 3: Bar plots of ATE IPTW estimate by different baselines.

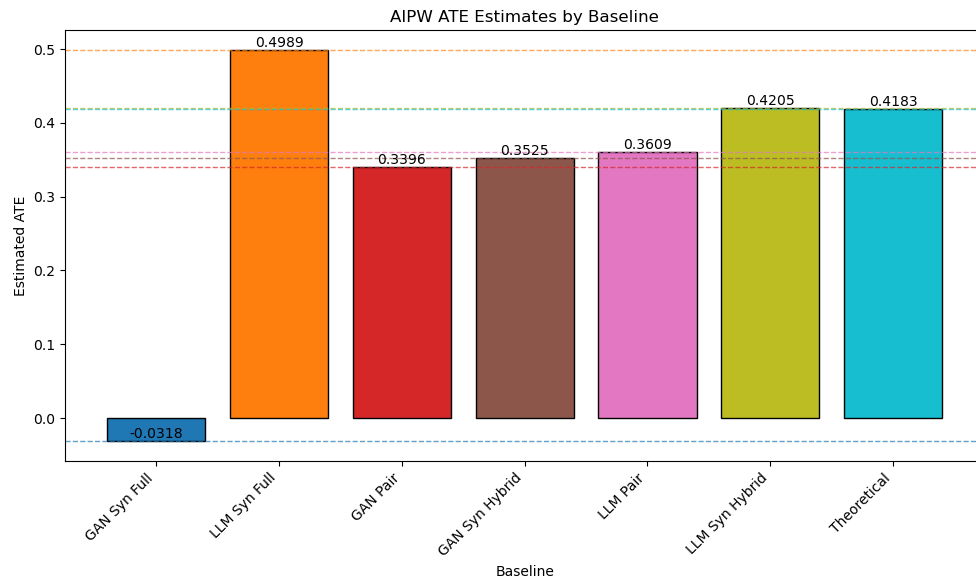


Figure 4: Bar plots of ATE AIPW estimate by different baselines.

that, in such scenarios, positivity violations may be more effectively addressed through sample augmentation rather than reweighting adjustments.

Table 2: IPTW and AIPW ATE Estimates and Their Absolute Differences from Theoretical ATE

Baseline	IPTW ATE	AIPW ATE	IPTW Diff	AIPW Diff
Raw	0.3169	0.1666	0.1014	0.2516
Raw (truncated)	0.3169	0.1666	0.1014	0.2516
GAN Pair	0.3861	0.3396	0.0321	0.0787
LLM Pair	0.4041	0.3622	0.0141	0.0561

6 LLM Synthetic data for realistic estimator evaluation

Practitioners often struggle to validate causal estimators on real-world data due to the absence of observable ground truth. While traditional simulation-based evaluations are widely used, they are often constrained by hand-crafted, oversimplified data-generating processes and fail to capture the complexity of real-world covariate structures — particularly in high-dimensional settings.

We introduce a novel evaluation framework that uses hybrid synthetic data to benchmark causal estimators under realistic and scalable conditions. Our method leverages LLM-generated covariates combined with separately estimated treatment and outcome models to create large-scale synthetic datasets that preserve both distributional and causal structure from real data. These synthetic datasets serve as high-fidelity “oracle” environments, enabling reliable ground-truth approximations of bias, variance, and mean squared error (MSE) for estimators such as IPTW, AIPW, and simple substitution.

Because the synthetic data is virtually unlimited, we can conduct controlled benchmarking across multiple replicates and conditions while maintaining real-world complexity. As shown in Algorithm 1, our hybrid generation protocol captures both covariate realism (via DCR) and key conditional dependencies ($A|W$, $Y|A, W$). Table 3 demonstrates that benchmarks based on LLM-hybrid synthetic data align more closely with the true estimator performance than those based on GANs or limited real-world subsets, highlighting the value of this protocol for trustworthy causal evaluation.

7 Real-World Applications and Limitations

We applied our method to a real-world dataset to illustrate a key limitation: without careful tuning, generative models can fail to capture essential structure, leading to synthetic data that misguides causal inference. Using NHANES data from two survey cycles (2011–2014), we built an observational dataset of 2,200 participants, including ten covariates, a binary treatment (“cvd”), and a binary outcome (“pfq051”).

As a basic validation, we trained classifiers to predict the outcome using (a) treatment, (b) covariates, and (c) both. Covariates were highly predictive, treatment alone had moderate signal, and the combination performed best. Some covariates—like age—strongly correlated with treatment. Importantly, propensity scores showed good overlap, supporting valid causal comparisons. Nonetheless, poor tuning of generative models undermined the synthetic data’s reliability for causal analysis.

As shown in Table 4, neither the LLM- nor GAN-based methods achieve a satisfactory AUC, suggesting possible issues with hyperparameter tuning. However, due to limited computational resources, we did not increase the number of training epochs. Despite the suboptimal distributional fidelity, the Hybrid method substantially improves AUC, potentially enhancing the estimation quality on GAN-generated synthetic data and aligning it more closely with the Data Seed benchmark. Notably, the estimated effect remains consistently small, providing no evidence of a significant treatment effect. Positivity violations are not a concern in this setting; hence, pairing does not offer additional benefits. This real-world scenario underscores the

Table 3: Oracle, Bias, Variance, and MSE for IPTW, AIPW, and Simple Substitution Estimators

IPTW Results				
Baseline	Oracle	Bias	Var	MSE
LLM	0.4295	0.0007	0.000648	0.000649
GAN	0.3538	0.0014	0.000795	0.000797
Truth	0.4172	-0.0019	0.000582	0.000585
AIPW Results				
Baseline	Oracle	Bias	Var	MSE
LLM	0.4204	-0.0170	0.000635	0.000923
GAN	0.3526	-0.0156	0.000739	0.000983
Truth	0.4137	-0.0221	0.000564	0.001053
Simple Substitution Results				
Baseline	Oracle	Bias	Var	MSE
LLM	0.4190	-0.0228	0.000627	0.001147
GAN	0.3511	-0.0207	0.000730	0.001157
Truth	0.4123	-0.0283	0.000559	0.001359

Table 4: Estimated ATE by IPTW, AIPW, and AUC on Test Set

Baseline	IPTW ATE	AIPW ATE	AUC
Data Seed	0.0120	0.0085	0.8505
Raw	0.0274	0.0044	–
LLM Syn Full	0.0075	0.0053	0.4496
LLM Syn Hybrid	–0.0004	0.0041	0.8024
LLM Pair	0.0274	0.0044	–
GAN Syn Full	0.0040	0.0035	0.5117
GAN Syn Hybrid	0.0056	0.0118	0.8045
GAN Pair	0.0274	0.0044	–
Raw (truncated)	0.0274	0.0044	–

importance of careful tuning in generative models to ensure even coarse alignment with the target data distribution.

References

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- Xu, L., Skoularidou, M., Cuesta-Infante, A., and Veeramachaneni, K. (2019). Modeling tabular data using conditional gan. In Wallach, H., Larochelle, H., Beygelzimer, A., d'Alché-Buc, F., Fox, E., and Garnett, R., editors, *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc.